

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A) | Date: 26-10-25
Time: 3 Hrs | Max. Marks: 300

26-10-25_SR.IIT_STAR CO-SC(MODEL-A)_JEE MAIN_GTM-2_SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

Rankers Academy JEE
THE NARAYANA GROUP
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PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

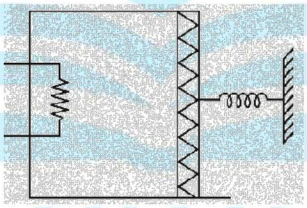
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A parallel plate capacitor has plates of area A separated by distance d between them. It is filled with a dielectric which has a dielectric constant that varies as $k(x) = K(1 + \alpha x)$ where x is the distance measured from one of the plates. If $(\alpha d) << 1$, the total capacitance of the system is best given by the expression

A) $\frac{AK\varepsilon_0}{d}\left(1+\frac{\alpha d}{2}\right)$ B) $\frac{AK\varepsilon_0}{d}\left(1+\left(\frac{\alpha d}{2}\right)^2\right)$ C) $\frac{A\varepsilon_0 K}{d}\left(1+\frac{\alpha^2 d^2}{2}\right)$ D) $\frac{AK\varepsilon_0}{d}(1+\alpha d)$

2. An ideal monoatomic gas is confined in a cylinder by a spring loaded piston of cross section 8.0×10^{-3} m^2. Initially the gas is at 300 K and occupies a volume of 2.4×10^{-3} m^3 and the spring is in its relaxed state as shown in figure. The gas is heated by a small heater until the piston moves out slowly by 0.2 m. The force constant of the spring is 8000 N/m and the atmospheric pressure is 1.0×10^5 N/m^2 . The cylinder and the piston are thermally insulated. The piston and the spring are massless and there is no friction between the piston and the cylinder. The final temperature of the gas will be: (Neglect the heat loss through the lead wires of the heater . The heat capacity of the heater coil is also negligible)



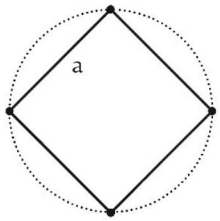
A) 300 B) 500 C) 800 D) 1500

3. Four identical hollow cylindrical columns of mild steel support a big structure of mass 50×10^5kg, The inner and outer radii of each column are 50cm and 100cm respectively. Assuming uniform local distribution, calculate the compression strain of each column. [Use $Y = 2.0 \times 10^{11}$ N/m^2, $g = 9.8$ m/s^2]

A) 3.60×10^{-2} B) 2.60×10^{-5} C) 1.87×10^{-5} D) 7.07×10^{-4}

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4. Four identical particles of mass 4M each are located at the corners of a square of side 'a'. What should be their speed if each of them revolves under the influence of other's gravitational field in a circular orbit circumscribing the square?



A) $4.05\sqrt{\frac{GM}{a}}$ B) $2.32\sqrt{\frac{GM}{a}}$ C) $4.96\sqrt{\frac{GM}{a}}$ D) $3.87\sqrt{\frac{GM}{a}}$

5. A series LR circuit is connected to an ac source of frequency ω and the inductive reactance is equal to R. A capacitance of capacitive reactance equal to R is added in series with L and R. The ratio of the new power factor to the old one is

A) $\sqrt{\frac{3}{2}}$ B) $\sqrt{\frac{2}{5}}$ C) $\frac{1}{\sqrt{2}}$ D) $\sqrt{2}$

6. The difference of speed of light in the two media A and B ($v_A - v_B$) is 5×10^7 m/s. If the refractive index of medium B is 1.5, then the ratio of refractive index of medium B to medium A is: (Given : speed of light in vacuum $c = 3 \times 10^8$ ms^{-1})

A) 1.30 B) 1.25 C) 1.20 D) 1.15

7. If A and B are two vectors satisfying the relation $A \cdot B = |A \times B|$. Then, the value of $|A+B|$ will be

A) $\sqrt{A^2+B^2}$ B) $\sqrt{A^2+B^2+\sqrt{2}AB}$ C) $\sqrt{A^2+B^2+2AB}$ D) $\sqrt{A^2+B^2-\sqrt{2}AB}$

8. A projectile fired at 30° to the ground is observed to be at same height at time 6s and 10s after projection, during its flight. The speed of projection of the projectile is ___ms^{-1}. (Given, $g = 10$ ms^{-2})

A) 120 B) 160 C) 140 D) 110

9. A stone of mass 4kg is tied to end of a massless string of length 1 m. If the breaking tension of the string is 400 N, then maximum linear velocity the stone can have without breaking the string while rotating in horizontal plane is

A) 10 ms^{-1} B) 40 ms^{-1} C) 400 ms^{-1} D) 20 ms^{-1}

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10. A thin circular disc of mass M and radius R is rotating in a horizontal plane about an axis passing through its centre and perpendicular to its plane with angular velocity ω. If another disc of same dimensions and mass M is placed gently on the first disc coaxially, then the new angular velocity of the system when they start rotating together is

A) $\frac{4}{5}\omega$ B) $\frac{3}{2}\omega$ C) $\frac{\omega}{2}$ D) $\frac{5}{4}\omega$

11. The weight of a body at the surface of earth is 54N. The weight of the body at an altitude of 3200 km above the earth's surface is (Given, radius of earth, R_e = 6400km).

A) 9.8 N B) 19.6 N C) 24 N D) 4.9 N

12. A solid metallic cube having total surface area 24 m^2 is uniformly heated. If its temperature is increased by 10°C, calculate the increase in volume of the cube. (Given coefficient of linear expansion $\alpha=2 \times 10^{-3}\,°C^{-1}$)

A) $2.4 \times 10^6\ cm^3$ B) $1.2 \times 10^5\ cm^3$ C) $6.0 \times 10^4\ cm^3$ D) $4.8 \times 10^5\ cm^3$

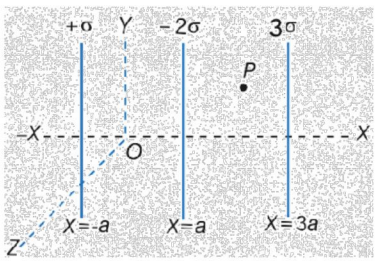
13. Under isothermal condition, the pressure of a gas is given by $P = \alpha V^{-2}$, where □ is a constant and V is the volume of the gas. The bulk modulus at constant temperature is equal to

A) 3P B) P/2 C) 2P D) P

14. The fundamental frequency of a closed organ pipe is equal to the first overtone frequency of an open organ pipe. If length of the open pipe is 120 cm, then the length of the closed pipe will be

A) 60 cm B) 30 cm C) 45 cm D) 15 cm

15. Three infinitely long charged thin sheets are placed as shown in figure. The magnitude of electric field at the point P is $x\sigma/\varepsilon_0$. The value of x is _____ .(all quantities are measured in SI units.)



A) 2 B) 3 C) 4 D) 5

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16. A rod of length 60 cm rotates with a uniform angular velocity 20 rad s^{-1} about its perpendicular bisector, in a uniform magnetic field 0.5 T. The direction of magnetic field is parallel to the axis of rotation. The potential difference between the two ends of the rod is ____ V.

A) 1.8 V B) 3.6 V C) 0 V D) 0.9 V

17. An alternating voltage V(t) = 220 sin 50πt volt is applied to a purely resistive load of 50Ω. The time taken for the current to rise from half of the peak value to the peak value is

A) 2.2 ms B) 5 ms C) 7.2 ms D) 6.6 ms

18. In Young's double slit experiment, light from two identical sources are superimposing on a screen. The path difference between the two lights reaching at a point on the screen is $\frac{3\lambda}{4}$. The ratio of intensity of fringe at this point with respect to the maximum intensity of the fringe is

A) $\frac{3}{4}$ B) $\frac{1}{3}$ C) $\frac{1}{2}$ D) $\frac{1}{4}$

19. When a metal surface is illuminated by light of wavelength λ the stopping potential is 8V. When the same surface is illuminated by light of wavelength 3λ, stopping potential is 2V. The threshold wavelength for this surface is

(A) 9λ B) 5λ C) 4.5λ D) 3λ

20. A sinusoidal voltage V(t) = 50 sin (500t) is applied across a pure inductance of L = 0.02 H. The current through the coil is

A) 5 cos(500t) B) –5 cos(500t) C) 5 sin(500t) D) –5 sin(500t)

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
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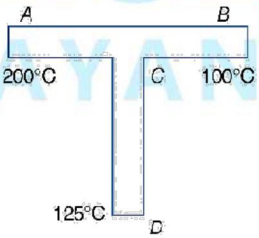
21. 4 moles of H_2 at 500 K is mixed with 2 moles of He at 400K. The mixture attains a temperature T and volume V. Now the mixture is compressed adiabatically to a volume V' and temperature T'. If $\frac{T'}{T}=\left(\frac{V}{V'}\right)^n$, find the value of 26n.

22. An alpha-particle of mass m suffers 1- dimensional elastic collision with a nucleus at rest of unknown mass. It is scattered directly backwards losing, 64% of its initial kinetic energy. The mass of the nucleus is □ × m. Then find □.

23. A particle initially at rest starts moving from reference point x =0 along X-axis, with velocity v that varies as, $v=2\sqrt{x}\ ms^{-1}$. The acceleration of the particle is _____ ms^{-2}.

24. Three identical spheres each of mass m are placed at the corners of a right angled triangle with mutually perpendicular sides equal to 2m each. Taking point of intersection of these two sides as origin, the magnitude of position vector of the centre of mass of the system is $\frac{4\sqrt{2}}{x}$, where the value of x is _____

25. A rod CD of thermal resistance 10.0 W/m×K is joined at the middle of an identical rod AB as shown in figure. The end A, B and D are maintained at 200°C, 100°C and 125°C, respectively. The heat current in CD is P W. The value of P is____.



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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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26. Which of the following is INCORRECT statement ?

A) In adiabatic expansion of an ideal gas, temperature of system always decrease

B) In isothermal expansion of an ideal gas, temperature of a gas always remains constant

C) In a reversible adiabatic compression of an ideal gas, change in entropy of system is zero

D) In an irreversible adiabatic compression of an ideal gas, change in entropy of system is positive

27. Which of the following arrangement does not represent correct increasing order of the property stated against it?

A) Sc Ti V Cr : Highest possible oxidation number

B) Ti V Cr Mn : Melting point

C) Cu Ni Co Fe : Paramagnetic behaviour

D) Mn Fe Co :$E°(M^{2+}/M)$standard reduction potential

28. Which of the following reaction is not correctly matched with their major product?

A) $CH_3CHCHCH_2CN \xrightarrow[(ii)H_2O]{(i)AlH(i-Bu)_2} CH_3CHCHCH_2CHO$

B) $CH_3CH_2CN \xrightarrow[(ii)H_2O/H^{\oplus}]{(i)SnCl_2/HCl} CH_3CH_2CHO$

C) $CH_3CH_2COCl + (CH_3)_2Cd \longrightarrow CH_3COCH_3 + CdCl_2$

D) $C_2H_5COCH_3 \xrightarrow{NaOCl} C_2H_5COO^-Na^+ + CHCl_3$

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29. Observe the following four solutions :

P : 10 ml, 0.1 M NaOH + 5ml, 0.1 M HCl

Q : 10 ml, 0.1 M NaOH + 15ml, 0.1 M CH_3COOH

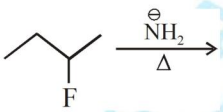
R : 10 ml, 0.1 M NH_3 + 10ml, 0.1 M NH_4Cl

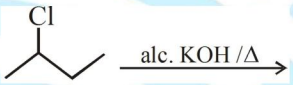
S : 10 ml, 0.05 M NaF + 5ml, 0.1 M HF

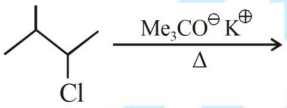
Which of above solutions act as buffer

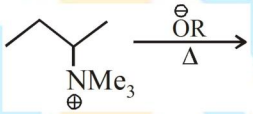
A) P and Q only B) R and S only C) P, R, S only D) Q, R, S only

30. Which of following reaction(s) produce Saytzeff product as a major product :

A) 

B) 

C) 

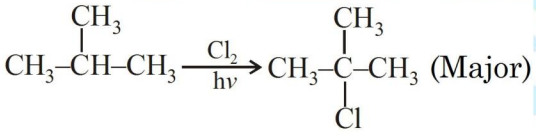
D) 

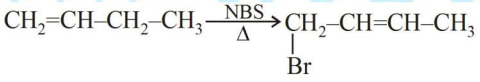
31. Which of the following liquid pair does not show contraction in volume on mixing-

A) $CHCl_3 + CH_3COCH_3$ B) $H_2O + HCl$

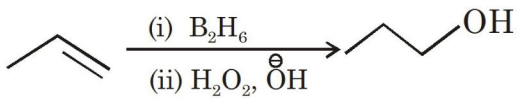
C) $H_2O + HNO_3$ D) $H_2O + C_2H_5OH$

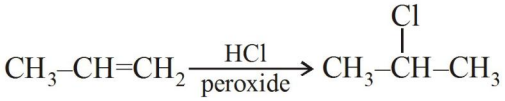
32. Select incorrect option:

A) 

B) 

(Major)

C) 

D) 

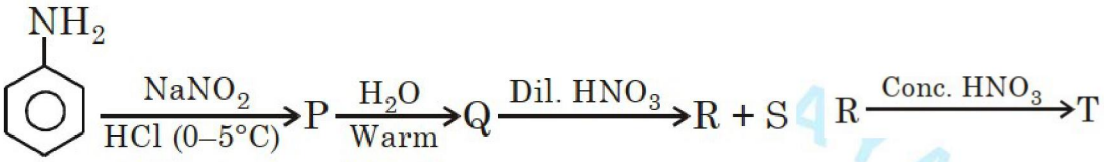
(Major)

For More Join: @IITJEE_Advanced

33. In the electrolysis of $CuSO_4$ solution, how many grams of Cu are plated out on the cathode in the time that it takes to liberate 5.6 litre of $O_2(g)$, measured at NTP at the anode?

A) 31.75 B) 14.2 C) 4.32 D) None of these

34. If R can form intramolecular hydrogen bonding & S can form intermolecular hydrogen bond. then correct pK_a order is :



A) T > S > R > Q B) Q > R > S > T C) T > R > S > Q D) R > S > T > Q

35. The entropy change for the formation of water vapour from 3.6 gm of liquid water at 100°C will be: (Molar heat of vapourisation of liquid water is 40.85 $kJmol^{-1}$)

A) 6.08 JK^{-1} B) 109.5 JK^{-1} C) 21.89 JK^{-1} D) –21.89 JK^{-1}

36. Arrange the following in increasing order of dipole moment:

NF_3, H_2S, HI, $CHCl_3$

A) HI < NF_3 < H_2S < $CHCl_3$ B) H_2S< NF_3 < $CHCl_3$ < HI

C) NF_3 < HI< H_2S < $CHCl_3$ D) HI < H_2S < NF_3 < $CHCl_3$

37. If Hund's rule is violated, then which among the following will become diamagnetic from paramagnetic

A) C_2 B) CN^- C) N_2^{-2} D) BN

38. The cell shown below generates a potential of 0.643 V at 300K. Find(log K_{sp}) for AgBr:

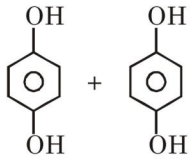
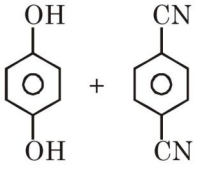
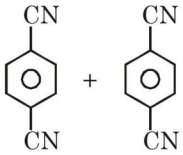
$Ag|AgBr(s)|NaBr(0.1M) \parallel AgNO_3(0.1M)|Ag$ $\left[\text{Take}\frac{2.303RT}{F}=0.06\right]$.

A) –30.2 B) –15.6 C) –12.71 D) –8.9

For More Join: @IITJEE_Advanced

39. Which of the following set contains specie that have interaction energy proportional to $\frac{1}{r^3}$?

($r \rightarrow$ internuclear distance)

A)  B)  C)  D) $KCl_{(aq.)} + NaCl_{(aq.)}$

40. Select the CORRECT statements?

(i) $[Zn(gly)_2]$ is optically active compound

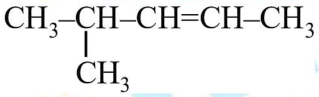
(ii) $[PdClBr(gly)]^-$ exhibits geometrical isomerism

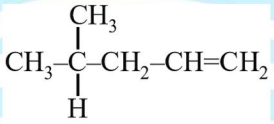
(iii) $[IrBr_4(en)]^-$ exhibits cis trans isomerism

(iv) $[RhCl_2(NH_3)_4]^+$ number of stereo isomers is equal to number of geometrical isomers

A) (i), (ii), (iv) B) (ii), (iii), (iv) C) (ii), (iii) D) (i), (iii), (iv)

41. $\underset{C_6H_{12}}{(X)} \xrightarrow[(ii)\ Zn/H_2O]{(i)O_3} (Y) + (Z)$, (Y) & (Z) both show disproportionation with conc. NaOH. Then X will be –

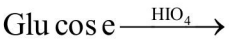
A)  B) $(CH_3)_3C–CH_2–CH=CH_2$

C) $(CH_3)_3C–CH=CH_2$ D) 

42. The CORRECT order of first ionization potential among the following is.

A) $F^- > Cl^-$ B) $Ga > Tl$ C) $Sn < Pb$ D) $O^- > S^-$

43. The products obtained are



A) HCOOH B) $CH_3–CHO$ C) HCHO D) Both (A) & (C)

For More Join: @IITJEE_Advanced

44. The correct order of increasing CFSE is :

A) $[PtCl_4]^{2-} < [PdCl_4]^{2-} < [NiCl_4]^{2-}$

B) $[Fe(H_2O)_6]^{3+} < [FeF_6]^{3-} < [Fe(CN)_6]^{3-}$

C) $[Ni(NH_3)_6]^{+2} < [Ni(en)_2(NH_3)_2]^{2+} < [Ni(en)_3]^{2+}$

D) $[NiCl_4]^{2-} < [Ni(CN)_4]^{2-} < [Ni(NH_3)_6]^{2+}$

45. Match List–I with List–II.

	List–I (Partial Derivatives)		List–II (Thermodynamic Quantity)
(a)	$\left(\frac{\partial G}{\partial T}\right)_P$	(i)	C_p
(b)	$\left(\frac{\partial H}{\partial T}\right)_P$	(ii)	$-S$
(c)	$\left(\frac{\partial G}{\partial P}\right)_T$	(iii)	C_v
(d)	$\left(\frac{\partial U}{\partial T}\right)_V$	(iv)	V

A) (a)–(ii), (b)–(i), (c)–(iii), (d)–(iv)

B) (a)–(i), (b)–(ii), (c)–(iv), (d)–(iii)

C) (a)–(ii), (b)–(i), (c)–(iv), (d)–(iii)

D) (a)–(ii), (b)–(iii), (c)–(i), (d)–(iv)

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
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46. The number of incorrect statement(s) from the following is –

(i) The successive half–lives of zero–order reactions decrease with time.

(ii) A substance appearing as reactant in the chemical equation may not affect the rate of reaction.

(iii) Order and molecularity of a chemical reaction can be a fractional number.

(iv) The rate–constant units of zero and second–order reactions are $mol\ L^{-1}s^{-1}$ and $mol^{-1}L\ s^{-1}$, respectively.

For More Join: @IITJEE_Advanced

47. If The enthalpy of formation of KCl(s) = x kcal mol^{-1}) from the data: Find |10X| ?

(i) $KOH(aq)+HCl(aq)\longrightarrow KCl(aq)+H_2O(l);\Delta H=-13.7\,kcal$

(ii) $H_2(g)+\frac{1}{2}O_2(g)\longrightarrow H_2O(l);\Delta H=-68.4\,kcal$

(iii) $\frac{1}{2}H_2(g)+\frac{1}{2}Cl_{(g)}+aq\longrightarrow HCl(aq);\Delta H=-39.3\,kcal$

(iv) $K(s)+\frac{1}{2}O_2(g)+\frac{1}{2}H_2(aq)+aq\longrightarrow KOH(aq);\Delta H=-116.5\,kcal$

(v) $KCl(s)+aq\longrightarrow KCl(aq);\Delta H=+4.4\,kcal$

48. Number of compounds that are colourless in their aqueous solution among the following is

$CuSO_4$, $ZnSO_4$, $FeSO_4$, $Cr_2(SO_4)_3$, NaCl, $KMnO_4$, $K_2Cr_2O_7$

49. How many of the following are not antiaromatic?

(i)  (ii)  (iii)  (iv) 

50. For a reaction of the type : A + B + C → Product ; the following observation are made ; doubling the conc. of A keeping the other constant doubles the rate ; doubling the concentration of B has no effect on the rate (keeping the other two concentration constant) & tripling the concentration of C (keeping that of the other two constant) increases the rate by a factor of 9. Find overall order of the reaction.

For More Join: @IITJEE_Advanced

MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If the solution curve of the differential equation $\frac{dy}{dx}=\frac{x+y-4}{x-y}$ passes through the point (3, 2) and (p+2, 3), p > 0, then

A) $2\tan^{-1}\left(\frac{1}{p}\right)=\log_e\left(p^2+1\right)$ B) $\tan^{-1}\left(\frac{1}{p}\right)=\log_e\left(p^2+1\right)$

C) $2\tan^{-1}\left(\frac{1}{p+1}\right)=\log_e\left(p^2+2p+2\right)$ D) $2\tan^{-1}\left(\frac{1}{p}\right)=\log_e\left(\frac{p^2+1}{p}\right)$

52. $\lim\limits_{x\to 1}\left(\sum\limits_{n=1}^{12}\frac{x}{n(n+1)x^2+3(2n+1)x+9}\right)$ is equal to

A) $\frac{11}{60}$ B) $\frac{3}{16}$ C) $\frac{13}{64}$ D) $\frac{9}{44}$

53. The area of the region bounded by $A=\{(x,y):x^2-4\le y\le \min\{2x+11,9,12-6x\}\}$ is

A) $\frac{511}{12}$ B) $\frac{947}{12}$ C) $\frac{437}{8}$ D) $\frac{240}{7}$

54. If function ' f ' satisfies $\int_0^1\left(f^4(x)-8x^2f^2(x)\right)dx+\frac{16}{5}=0$, then the value of $f^2(1)$ is

A) 0 B) 1 C) 4 D) 2

55. For polynomial $P(x)=1-\frac{1}{3}x+\frac{1}{6}x^2$, define $Q(x)=P(x)\cdot P(x^3)\cdot P(x^5)\cdot P(x^7)\cdot P(x^9)=\sum\limits_{i=0}^{50}a_ix^i$, then $\sum\limits_{i=0}^{50}|a_i|=$ _____

A) $\frac{275}{37}$ B) $\frac{243}{32}$ C) $\frac{243}{39}$ D) $\frac{245}{37}$

56. If one side of square ABCD is on the line y = 2x – 17 and the other two vertices lies on the parabola $y = x^2$, then the minimum area of the square is

A) 160 B) 1200 C) 80 D) 60

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57. The value of the definite integral $\int_0^2 \left(\sqrt{1+x^3}+\sqrt[3]{x^2+2x}\right)dx$ is

A) 4 B) 5 C) 6 D) 7

58. $\int \frac{x^{14}+x^{11}+x^5}{\left(x^6+x^3+1\right)^3}dx = f(x)+C$ and $f(-1)=\frac{1}{3}$, then the value of $\frac{27}{8}f(1)$ is equal to

A) $\frac{1}{81}$ B) 0 C) $\frac{1}{8}$ D) $-\frac{1}{8}$

59. The number of points of intersection of the curves represented by $\arg(z-2-7i)=\cot^{-1}(2)$ and $\arg\left(\frac{z-5i}{z+2-i}\right)=\pm\frac{\pi}{2}$

A) 0 B) 1 C) 2 D) 4

60. Consider the following frequency distribution

Class	0 – 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
Frequency	3	a	b	11	8	4

If the mean is $\frac{277}{9}$ and median is $\frac{335}{11}$, then the value of 4a + b is

A) 40 B) 38 C) 50 D) 48

61. Let $b_0=2, b_1=5, 2b_{n+1}=b_n^2+1\ \forall n\geq 1$, then the value of $\frac{1}{b_0+1}+\frac{1}{b_1+1}+\frac{1}{b_2+1}...+\frac{1}{b_n+1}+\frac{1}{b_{n+1}-1}$ is equal to

A) 1 B) $\frac{1}{3}$ C) $\frac{1}{2}$ D) $\frac{7}{12}$

62. If true range of function $f(x)=1+\cos x+\cos^2 x\ \ \forall x\in R$ is [a, b] then $\frac{b}{3}$ is

A) –1 B) 1 C) $\frac{1}{2}$ D) $\frac{1}{3}$

63. If a function $f(x) = (a-2)x^3 + 3x^2 - bx$ is symmetrical about line x = 3 and local minimum value of f(x) is 'c' then value of a + 3b +2c is equal to

A) 3 B) 4 C) 5 D) 2

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64. If $\bar{a},\bar{b},\bar{c},\bar{d}$ are four co-planar points, then

A) $[\bar{a}\ \bar{b}\ \bar{c}]+[\bar{a}\ \bar{c}\ \bar{d}]+[\bar{a}\ \bar{d}\ \bar{b}]=[\bar{b}\ \bar{c}\ \bar{d}]$

B) $[\bar{a}\ \bar{b}\ \bar{c}]+[\bar{a}\ \bar{c}\ \bar{d}]+[\bar{a}\ \bar{d}\ \bar{b}]=[\bar{b}\ \bar{c}\ \bar{a}]$

C) $[\bar{a}\ \bar{b}\ \bar{c}]+[\bar{a}\ \bar{c}\ \bar{d}]+[\bar{a}\ \bar{d}\ \bar{b}]=[\bar{c}\ \bar{d}\ \bar{a}]$

D) $[\bar{a}\ \bar{b}\ \bar{c}]+[\bar{a}\ \bar{c}\ \bar{d}]+[\bar{a}\ \bar{d}\ \bar{b}]=[\overline{2a}\ \bar{c}\ \bar{d}]$

65. Let a cube painted red in all 6 faces and cut into 125 small equal cubes. If one cube is picked at random out of 125 small cubes and at least one face found to be red, then probability of exactly 3 faces showing red is p. The value of $\frac{49}{60}P$ is

A) $\frac{4}{75}$ B) $\frac{4}{49}$ C) $\frac{1}{8}$ D) $\frac{1}{15}$

66. Number of ways to score exactly '8' runs in an over, when batsmen don't hit '4' in any boll,(considering no extra ball), is

A) ${}^{13}C_8-6{}^8C_4$ B) ${}^{13}C_8-6{}^8C_4-21$ C) ${}^{13}C_8-6{}^9C_4-35$ D) ${}^{13}C_8-6{}^9C_4+35$

67. If $\frac{1}{\sin 2^\circ}+\frac{1}{\sin 4^\circ}+\frac{1}{\sin 8^\circ}+\ldots\frac{1}{\sin 128^\circ}$ is $\frac{\sin A}{\sin B\sin C}$, where A, B, C are acute angle (In degrees), then

A) A, B, C is A.P B) A = B + C C) A = |B – C| D) B = A+ C

68. The number of straight lines passing through the point (3, 5) and forming a triangle of area 18 square units with the co-ordinate axes, is/are

A) 0 B) 2 C) 4 D) Infinite

69. The range of 'r' so that the circles $(x-2)^2+(y-4)^2=r^2$ and $(x-5)^2+(y-1)^2=16$ intersect at two distinct points, is

A) $r\in\left(3\sqrt{2}-4,4+3\sqrt{2}\right)$ B) $r\in\left(4-2\sqrt{2},4+\sqrt{2}\right)$

C) $r\in\left(4-\sqrt{2},4+2\sqrt{2}\right)$ D) $\left(4\sqrt{2}-3,4+3\sqrt{2}\right)$

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70. If the roots of the equation $ax^2 + bx + c = 0$ are α and β, then the roots of the equation $a(3x + 1)^2 - b(3x + 1)(x - 1) + c(x - 1)^2 = 0$ are

A) $\frac{3\alpha+1}{\alpha-1}, \frac{3\beta+1}{\beta-1}$ B) $\frac{\alpha+1}{\alpha-3}, \frac{\beta+1}{\beta-3}$ C) $\frac{\alpha-1}{\alpha+3}, \frac{\beta-1}{\beta+3}$ D) $\frac{3\alpha+1}{1-\alpha}, \frac{3\beta+1}{1-\beta}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. The value of $\sum_{k=22}^{66} \frac{1}{\sin(2k+1)^\circ \sin(2k+2)^\circ} = \sec(\lambda^\circ)$, where λ is an acute angle, then the value of λ is

72. The integral $16\int_1^2 \frac{dx}{x^3(x^2+2)^2\,dx} = \frac{a}{b} - \ln 4$, where a & b are coprime, the value of (a – b) equals

73. Consider a function $f(x) = \prod_{r=1}^{\infty}\left(\frac{1+2\cos\left(\frac{2x}{3^r}\right)}{3}\right)$. Then the number of points where $g(x) = |xf(x)| + ||x-2|-1|$ is non differentiable on interval $(0, 3\pi)$, is/are

74. For the hyperbola $\frac{x^2}{25} - \frac{y^2}{45} = 1$, the chord $x\cos\theta + y\sin\theta = \lambda$ subtends a right angle at the centre, and the length of tangent to the circle, concentric with this hyperbola, to which the given chord is a tangent, from the point $P\left(5, 5\sqrt{2}\right)$ is l, then the value of $4l^2$ is equal to

75. If the angle between lines belonging to the family of lines $(3x+y-7)+\lambda(x+y-3)=0$ and $(x+3y-7)+\mu(x+4y-3)=0$ which are farthest from origin (λ and μ are real parameters), is $\tan^{-1}\left(\frac{a}{b}\right)$ where a, b are coprime natural numbers, then b – a is

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